

COMPARISON OF SQL AND NOSQL DATABASES

1. Introduction

Databases are used to store, manage, and retrieve data efficiently. **SQL databases** use a relational model based on tables, rows, columns, and relationships. **NoSQL databases** use flexible data models such as documents, key-value pairs, graphs, or wide-column stores.

For the E-Commerce Order Management System, SQL is suitable for managing structured data such as customers, products, orders, payments, and shipping, while NoSQL can be useful for flexible data such as a product catalog represented using JSON documents.

2. Comparison Between SQL and NoSQL

Feature	SQL Database	NoSQL Database
Data Model	Tables and relations	Documents, key-value, graph, or column-based
Schema	Fixed and predefined	Flexible or schema-less
Data Format	Rows and columns	JSON documents, key-value pairs, etc.
Relationships	Primary and foreign keys	Embedded or referenced data
Query Language	SQL	Database-specific query methods
Normalization	Commonly normalized	Denormalization is often used
Scalability	Primarily vertical scaling	Often designed for horizontal scaling
Transactions	Strong ACID support	Depends on the database
Best for	Structured and related data	Flexible or rapidly changing data

3. Advantages of SQL Databases

- **Data integrity:** Primary keys, foreign keys, and constraints maintain accurate and consistent data.
- **ACID transactions:** Suitable for applications where reliable transactions are important.

- **Complex queries:** Supports joins, subqueries, aggregate functions, and advanced SQL queries.
- **Structured data:** Best suited for well-organized data with clearly defined relationships.
- **Standardized language:** SQL is widely used across relational database systems.

Common Use Cases

- Banking and financial systems
 - E-Commerce order and payment management
 - Student and employee management systems
 - ERP and CRM systems
 - Inventory management systems
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4. Advantages of NoSQL Databases

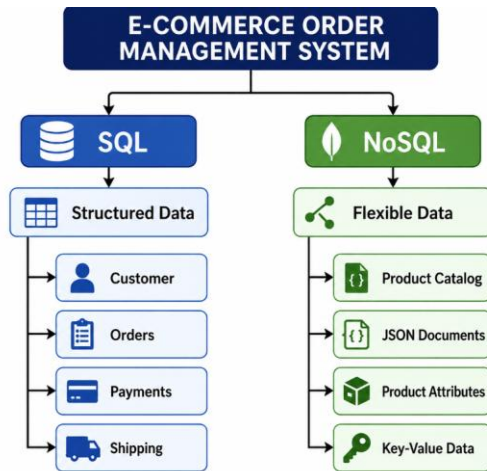
- **Flexible schema:** New attributes can be added easily.
- **Scalability:** Suitable for handling large volumes of data.
- **High performance:** Useful for applications requiring fast data access.
- **Supports different data types:** Can store structured, semi-structured, and unstructured data.
- **Multiple data models:** Supports document, key-value, graph, and wide-column models.

Common Use Cases

- Social media applications
 - Real-time web applications
 - Big data and analytics
 - Content management systems
 - IoT applications
 - Product catalogs with changing attributes
 - Gaming and mobile applications
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5. SQL vs NoSQL for the E-Commerce System

Your project can use both approaches for different purposes:



SQL Example

PRODUCT

Product_ID

Product_Name

Category_ID

Seller_ID

Price

Stock_Quantity

Product_Status

NoSQL Example

```
{
  "product_id": 401,
  "product_name": "Wireless Mouse",
  "category": "Electronics",
  "seller": "Tech World",
  "price": 799.00,
  "stock_quantity": 50,
```

```
"product_status": "Available"  
}
```

6. Conclusion

Both SQL and NoSQL databases have important advantages and are suitable for different requirements.

SQL databases are best suited for structured data, complex relationships, transactions, and strong data consistency. Therefore, SQL is highly suitable for the main **E-Commerce Order Management System**, including customers, orders, payments, and shipping.

NoSQL databases are useful when data is flexible, rapidly changing, or large in volume. For this project, a NoSQL document model can be used to represent the **Product Catalog** using JSON documents and key-value structures.

Final Statement

“SQL is preferred for structured and transactional data, whereas NoSQL is preferred for flexible, scalable, and semi-structured data. The choice depends on the application's requirements.”